

Pulse Sequence Sensing and Pulse Position Modulation for Optical Integrated Sensing and Communication

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Abstract—The future wireless network is expected to provide communication and sensing abilities simultaneously, and the optical spectrum brings a promising candidate for integrated sensing and communication. In this letter, an integrated sensing and communication scheme based on pulse sequence sensing and pulse position modulation is proposed for free space optics. By generating a unipolar optical signal, optical wireless communication and optical sensing can adopt the intensity modulation and direct detection scheme simultaneously. Benefited from the nearly-ideal correlation properties of the pulse sequence, the communication receiver can detect the pulse position, and the sensing receiver can estimate the target distance. Moreover, different performance metrics of communication and sensing are derived theoretically. Simulations show that the proposed method can carry out communication and sensing simultaneously, even in the multi-user scenario. Furthermore, different optical systems can adjust system parameters to adapt to different scenarios, so that they will enhance the ability of the future wireless network.

Index Terms—Integrated sensing and communication, optical wireless communication, optical sensing.

I. INTRODUCTION

RECENTLY, the concept of integrated sensing and communication (ISAC) has received considerable attention from both academia and industry. A radio frequency (RF) based ISAC system can utilize the similarities in hardware architectures and signal processing techniques between communication and sensing, to achieve spectrum sharing [1] and hardware reuse [2]. Moreover, the future wireless network is expected to simultaneously provide large communication capacity and accurate sensing ability of the environment, which is also known as the ISAC system [3]. Therefore,

numerous research has been conducted on RF-based ISAC, including waveform design [4], network design [5], and resource management [6].

Compared with the emerging research on RF-based ISAC, much less research is conducted on optical ISAC. However, since the optical spectrum has a much larger bandwidth than its RF counterpart, it can provide high-speed, short-range communication ability. With the advantages of large capacity, high security, and license-free spectrum, optical wireless communication (OWC) can operate in both indoor and outdoor scenarios [7]. Meanwhile, optical sensing devices like laser radars provide non-contact methods for target localization [8]. Furthermore, the intensity modulation and direct detection (IM/DD) scheme is frequently adopted by both OWC and optical sensing devices [9]. Therefore, integrating OWC and optical sensing in a single system becomes a promising trend.

Some studies adopted visible light communication (VLC) for optical ISAC. Béchadargue et al. suggested using the headlamps and taillights of vehicles to perform simultaneously communication and range-finding [10]. Similarly, a VLC-based vehicle localization method was developed in [11]. As for ISAC based on laser radar, Mizui et al. proposed a technique called boomerang transmission, where two vehicles alternately relay data multiplied by pseudo-noise (PN) codes [12]. The boomerang transmission system, which was further improved by [13], could provide active safety and smooth driving abilities to autonomous vehicles. However, most of these methods should work with cooperative targets, i.e., objects equipped with ISAC transceivers, where the reflected optical signal is not fully utilized.

In this letter, we propose a pulse sequence sensing and pulse position modulation (PSS-PPM) method for optical ISAC, which provides sensing and communication abilities for a laser radar simultaneously. Pulse position modulation (PPM) has a higher power efficiency compared to that of on-off keying, which improves the endurance of an optical system. The spread spectrum technique adopted by [12] is further exploited to reduce multi-user interference (MUI). More importantly, since the proposed method utilizes optical signals reflected by targets, it can work with not only ISAC transceivers but also non-cooperative targets, which makes PSS-PPM flexible for both ISAC scenarios and conventional sensing tasks.

The remainder of this letter is organized as follows. In Section II, the signal structure and system model are introduced. Section III explains the proposed demodulation and ranging algorithms in detail, and derives different performance metrics of the proposed method. The numerical results of communication and sensing performances are illustrated in Section IV. Finally, the conclusion is drawn in Section V.

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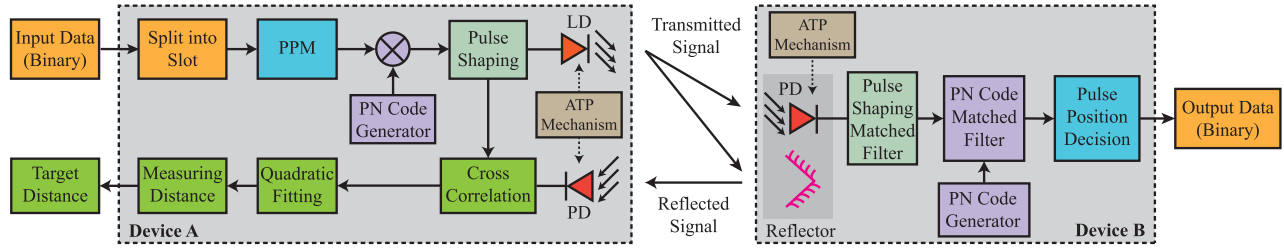


Fig. 1. System model of the proposed PSS-PPM method for optical ISAC.

II. SIGNAL STRUCTURE AND SYSTEM MODEL

In this section, we introduce the system model and the signal structure of the proposed PSS-PPM method. For simplicity, the system model illustrates a uni-directional communication and sensing scenario. By means of time division multiplexing or wavelength division multiplexing, the model can also be extended to bi-directional communication and sensing. Fig. 1 shows the basic system model that implements the PSS-PPM method, and Fig. 2 depicts the signal structure in one slot.

A. System Model

Device A acts as a communication transmitter and a sensing transceiver simultaneously in the ISAC scenario. To establish the communication link to Device B, Device A may contain acquisition, tracking, and pointing (ATP) mechanisms. With the assistance of beacon lights, computer vision methods or RF communication, a line-of-sight link between Device A and Device B can be established. Then, Device A can communicate with Device B and estimate the distance between them simultaneously.

In Device A, the input communication data is split into slots and transformed into pulse positions. The PN code allocated to Device A is modulated so that it starts at the aforementioned pulse positions. A pulse shaping filter is then applied to the PN code to generate the PSS-PPM signal. Finally, the PSS-PPM signal is transmitted to free space by a laser diode (LD).

Once a link exists for the communication receiver, Device B utilizes a photodiode (PD) to detect the optical signal transmitted by Device A. The outputs of the pulse shaping matched filter represent the received PSS-PPM symbols. Then, the symbols come into the ensuing PN code matched filter, and Device B demodulates the data by pulse position decision.

Meanwhile, dihedral corner reflectors are mounted within the beam spot near the PD of Device B, so that the reflected beam can reach the PD of Device A. Then, Device A can calculate the cross-correlation between its transmitted signal and its received signal. By detecting the maximum value in the cross-correlation result, Device A gets a maximum-likelihood estimation (MLE) of the time delay. Then, a quadratic fitting method is further adopted to obtain a more precise estimation of the target distance.

B. Signal Structure

Slot is the fundamental signal structure in the proposed PSS-PPM method. Each slot has a duration of T_s , and the pulse repetition frequency (PRF) of the laser radar is $1/T_s$. A slot is split into a PPM interval with a duration of T_p and a guard

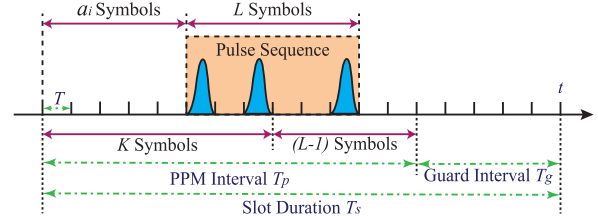


Fig. 2. Signal structure of the proposed PSS-PPM method for optical ISAC.

interval with a duration of T_g , in order to accommodate the requirement of unambiguous range D_u .

The PPM interval contains $(K + L - 1)$ symbols, where K and L denote the PPM modulation order and the amount of symbols in each pulse sequence, respectively. Each symbol has a duration of T and contains a unipolar pulse $g(t)$. The input communication bit stream is split into slots with $\log_2 K$ bits in each slot, and the i -th slot is transformed into pulse position $a_i \in \{0, 1, \dots, K - 1\}$. For the i -th slot, the pulse sequence starts at the a_i -th symbol of the PPM interval and lasts for L consecutive symbols. Meanwhile, no pulse is transmitted in the guard interval.

The pulse $g(t)$ is generated by a pulse shaping filter, and the Gaussian pulse is usually adopted by laser radars. However, the Gaussian pulse has a complicated mathematical expression and an infinite pulse length. Instead, a unipolar raised-cosine pulse is adopted as

$$g(t) = \begin{cases} \frac{1}{2} \left(1 + \cos \left(\frac{2\pi}{T} \left(t - \frac{T}{2} \right) \right) \right), & 0 \leq t \leq T, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

The pulse $g(t)$ is further modulated by a unipolar pseudo noise (PN) code $c_0[j] \in \{0, 1\}$, $0 \leq j < L$ to generate a pulse sequence with ideal correlation properties. Unfortunately, PN codes with a finite length cannot have ideal cross-correlation and auto-correlation properties simultaneously [14]. Nevertheless, with sufficiently large L , code sets like m-sequences can have nearly ideal correlation properties, so m-sequences are adopted by the PSS-PPM method.

Supposing that the pulse positions to be transmitted are $\{a_i\}$, the transmitted signal is expressed as

$$s_T(t) = \sum_{i=0}^{N_s-1} \sum_{j=0}^{L-1} c_0[j] g(t - iT_s - (a_i + j)T). \quad (2)$$

When N_u devices are working together, the received signal of both Device A and Device B can be expressed as

$$s_R(t) = h_0(t) s_T(t - \tau_0) + \sum_{m=1}^{N_u-1} h_m(t) s_m(t - \tau_m) + n(t), \quad (3)$$

where $s_T(t)$ is the desired signal. In Fig. 1, the desired signal for Device B is the signal transmitted by Device A, and the desired signal for Device A is the reflected signal transmitted by itself. Besides, $s_m(t)$ and τ_m ($1 \leq m < N_u$) are the m -th interference signal and its time delay, respectively. $n(t) \sim \mathcal{N}(0, \sigma_w^2)$ denotes the additive white Gaussian noise (AWGN). $h_m(t)$ ($0 \leq m < N_u$) depicts the influences of atmospheric losses and turbulence, which can be described by statistical models [15]. The index $m = 0$ and $1 \leq m < N_u$ correspond to the channel of the desired signal and the interference signal, respectively.

III. THEORETICAL PERFORMANCE METRICS

In this section, we derive different performance metrics of communication and sensing. The achievable data rate and BER indicate the efficiency and reliability of the communication system, respectively. Meanwhile, unambiguous range, MSE and resolution indicate the ability of accurate distance measurement for optical sensing. The influences of MUI are also analyzed with the method of Gaussian approximation.

A. Communication Metrics

1) *Achievable Rate*: According to Section II-B, each slot transmits $\log_2 K$ bits. Meanwhile, the PPM modulation order K also affects the duration of a slot, i.e., $T_s = T_g + (K + L - 1)T$. Thus, the achievable rate is given by

$$C(K) = \frac{\log_2 K}{T_g + (K + L - 1)T}. \quad (4)$$

If T_g , L , and T are fixed, $C(K)$ is a unimodal function of K . Let $dC(K)/dK = 0$, and the optimal modulation order K_0 to maximize the achievable rate can be obtained by

$$K_0 = \exp \left(W \left(\frac{1}{e} \left(L - 1 + \frac{T_g}{T} \right) \right) + 1 \right), \quad (5)$$

where e denotes Euler number, and $W(\cdot)$ denotes Lambert W-function, i.e., $W^{-1}(x) = x \exp(x)$.

2) *BER*: For the communication receiver, supposing that the received signal is well synchronized, the received PSS-PPM symbols can be obtained from the output of the pulse shaping matched filter. The j -th symbol in the i -th slot is expressed as

$$r_i[j] = \int_0^T s_R(t + iT_s + jT + \tau_0) g(t) dt. \quad (6)$$

Since $s_T(t)$ is unipolar, the communication receiver utilizes bipolar reference symbols $p_j[m]$ for the PN code matched filter to achieve a superior auto-correlation performance. The reference symbols are given by

$$p_j[m] = \begin{cases} 2c_0[m - j] - 1, & j \leq m \leq j + L - 1, \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

If the PN codes of $s_R(t)$ and $c_0[j]$ are the same, a sharp peak occurs at the pulse position a_i . The MLE for pulse position detection is represented as

$$\hat{a}_i = \arg \max_{0 \leq j < K} \sum_{m=0}^{K+L-2} r_i[m] p_j[m]. \quad (8)$$

To obtain the bound of BER, we define an integral kernel as

$$\varphi(z, I) = \left(\phi \left(z + I \sqrt{\frac{T_s(L+1-\delta(L))E_s}{4TLN_0}} \right) \right)^{K-1}, \quad (9)$$

where $\phi(\cdot)$ denotes the cumulative distribution function of standard Gaussian distribution. The variable z denotes the normalized amplitude of the correlation peak, and the correction term $\delta(L)$ describes the non-ideal correlation properties of m-sequences. The lower and upper bounds of BER correspond to $\delta(L) = 0$ and $\delta(L) = \sqrt{L}$, respectively. The average symbol SNR of the received signal is expressed as

$$\frac{E_s}{N_0} = \frac{3(L+1)T}{8T_s\sigma_w^2}. \quad (10)$$

The influences of atmospheric losses and turbulence are included in a random variable $I = \int_0^{T_s} h_0(t) dt / T_s$, whose probability density function is $p(I)$. Then, the symbol error rate (SER) for communication is derived as

$$P_{es} = 1 - \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{p(I)}{\sqrt{2\pi}} \exp \left(-\frac{z^2}{2} \right) \varphi(z, I) dz dI. \quad (11)$$

If the transmitted pulse positions $\{a_i\}$ are randomly selected from $\{0, 1, \dots, K-1\}$ with an equal probability, the relationship between BER and SER is established as

$$P_{eb} = \frac{K}{2(K-1)} P_{es}. \quad (12)$$

Supposing that $s_R(t)$ is sampled at the rate of R_s , the complexity of calculating product and sum is $\mathcal{O}((K+L)R_sT)$ for the pulse shaping matched filter and $\mathcal{O}(K(K+L))$ for pulse position detection in one slot. Thus, the total complexity in one slot is $\mathcal{O}((K+L)(R_sT + K))$.

B. Sensing Metrics

1) *Unambiguous Range*: The reflected pulse sequence should arrive at the receiver in the same slot to avoid range ambiguities. Thus, the constraint on the guard interval is given by $T_g = 2D_u/c$, where c is the speed of light.

2) *MSE*: MSE for sensing comes from the noise and interference in $s_R(t)$. By calculating the cross-correlation between $s_T(t)$ and $s_R(t)$, the sensing receiver gets an MLE of the target distance D as

$$\hat{D} = \frac{c}{2} \arg \max_{\tau_0} \int_0^{T_s} s_R(t) s_T(t - \tau_0) dt. \quad (13)$$

The Cramér-Rao Bound (CRB) is the lower bound of MSE for an unbiased estimator, which depicts the sensing

performance of MLE with enough sampling points. If $s_R(t)$ is sampled at the rate of R_s , the CRB is expressed as

$$\varepsilon_D^2 = \mathbb{E} \left(\left(\hat{D} - D \right)^2 \right) \geq \frac{c^2}{4} \cdot \frac{3T^2}{2\pi^2 T_s R_s} \cdot \frac{N_0}{E_s}, \quad (14)$$

where $\mathbb{E}(\cdot)$ means calculating expectation [16].

3) *Resolution*: Compared with MSE, the resolution ΔD depicts the error brought by the non-ideal laser radar pulse, which is inversely proportional to the signal bandwidth. The bandwidth of a raised-cosine pulse in (1) is $B = 2/T$, so the resolution is expressed as $\Delta D_B = c/2B = cT/4$.

The conversion from analog signal to digital signal also affects the resolution. If MLE merely gives an estimation of integer sampling points, the sampled resolution is given by $\Delta D_s = c/4R_s$.

Moreover, the quadratic fitting method [16] is adopted to break the limitation of sampling. Utilizing the similarity between a raised-cosine pulse and a quadratic function near the axis of symmetry, the quadratic fitting method uses least square approximation to estimate the fractional part of $\tau_0 R_s$.

To attain an estimation $\hat{\tau}_0$ that approaches CRB in the high-SNR region, both the MLE and the quadratic fitting should be utilized to estimate the integer part and the fractional part of $\tau_0 R_s$, respectively. The complexity of the MLE is $\mathcal{O}(T_s R_s \log(T_s R_s))$ if FFT is utilized, while the complexity of the quadratic fitting is $\mathcal{O}(N_f)$ if N_f sampled points are adopted in the least square approximation. Therefore, the total complexity in one slot is $\mathcal{O}(T_s R_s \log(T_s R_s) + N_f)$.

C. Influences of MUI

In the derivation of BER for communication and MSE for sensing, MUI can be modelled as a zero mean Gaussian random process to get more accurate results [17]. The auto-correlation of $g(t)$ and the cross-correlation between $(2c_0[j] - 1)$ and $c_m[j]$ are defined as $R_{gg}(t)$ and $\tilde{R}_{cc}[j]$, respectively. Afterwards, the equivalent noise power caused by the m -th interference can be approximated as

$$\sigma_m^2 \approx \frac{3}{T_s} \sigma \left(R_{gg}(\delta_m T) \tilde{R}_{cc}[q_m] + R_{gg}(T(1 - \delta_m)) \tilde{R}_{cc}[q_m - 1] \right), \quad (15)$$

where δ_m and q_m are the fractional part and the integer part of $(\tau_m - \tau_0)/T$, respectively. The notation $\sigma(\cdot)$ means calculating standard deviation for δ_m and q_m .

Hence, the equivalent noise plus interference power is approximated as

$$\tilde{\sigma}^2 \approx \sigma_w^2 + \sum_{m=1}^{N_u-1} \sigma_m^2. \quad (16)$$

By substituting σ_w^2 with $\tilde{\sigma}^2$ in (10), the performance metrics can be derived in the multi-user scenario.

IV. NUMERICAL RESULTS

Table I shows the parameters for numerical simulations of the proposed PSS-PPM method. 10^6 Monte Carlo experiments are carried out to calculate BER for communication and MSE

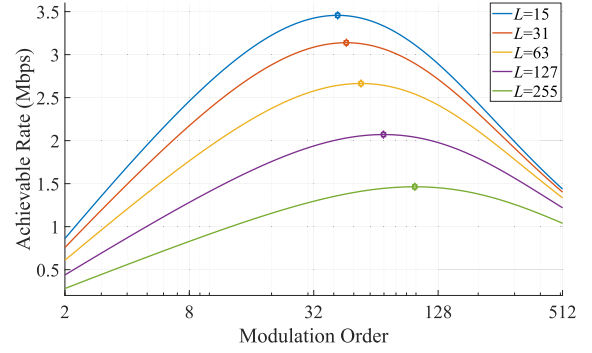


Fig. 3. Achievable rate with respect to the modulation order and the length of PN codes. The markers correspond to the optimal modulation order K_0 .

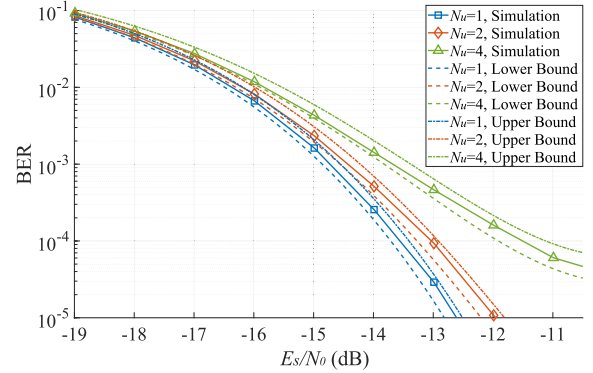


Fig. 4. BER for communication with respect to the number of users in the presence of atmospheric losses and turbulence.

TABLE I
SIMULATION CONFIGURATIONS

Parameter	Symbol	Value
Unambiguous range	D_u	300 m
Rytov variance	σ_R^2	0.04
Guard interval	T_g	2 μ s
Symbol duration	T	10 ns
Slot duration	T_s	3.9 μ s
Sampling rate	R_s	1 GHz
Number of users	N_u	1, 2, 4

for sensing. The simulations are carried out on the condition of $K = 64$ and $L = 127$, except for Fig. 3, which shows the achievable rate $C(K)$ with respect to the modulation order K and the length L of PN codes. The $\log_2 K$ binary numbers in each slot are transformed into decimals to get the pulse positions, so that $\{a_i\}$ are randomly selected from $\{0, 1, \dots, K-1\}$ with an equal probability. Time delay τ_0 of $s_T(t)$ is uniformly distributed on the interval $[0, T_g]$, while τ_m of $s_m(t)$ is uniformly distributed on the interval $[-T_s, T_s]$. The mean intensity of each $s_m(t)$ is the same as that of $s_T(t)$. $h_m(t)$ ($0 \leq m < N_u$) follows log-normal distribution, whose mean and variance can be calculated by Beers-Lambert Law and Rytov approximation, respectively, which can be further adopted to evaluate the influences of atmospheric losses and turbulence in the simulations [15].

In Fig. 3, the duration T of each symbol and the duration T_g of the guard interval are fixed, and the duration T_s of a slot varies with different K and L . Due to the unimodality of the achievable rate $C(K)$, a unique optimal modulation order K_0 exists to maximize $C(K)$. According to Table II,

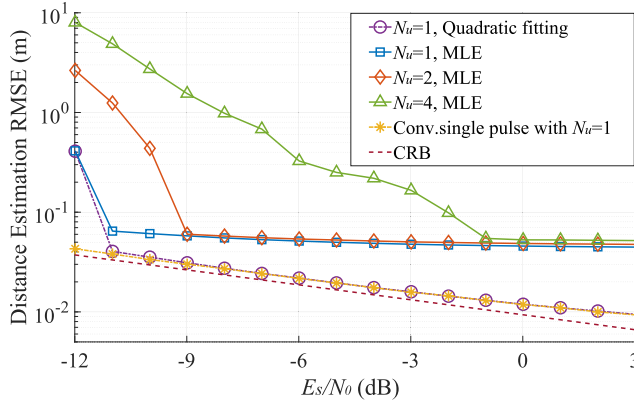


Fig. 5. RMSE for sensing with respect to the number of users in the presence of atmospheric losses and turbulence.

TABLE II
OPTIMAL MODULATION ORDER K_0

Length of PN codes	15	31	63	127	255
Theoretical results	66.83	70.61	78.04	92.44	119.90
Numerical results	67	71	78	92	120

the theoretical values of K_0 obtained from (5) correspond with the numerical results well.

Fig. 4 shows BER for communication with respect to the number of users. By modelling MUI as a zero-mean Gaussian process, the BER performance in the multi-user scenario can be described. Due to the non-ideal correlation properties of m-sequences and the asynchronous interference signal, gaps exist between the simulation results and the lower bounds, which can only be achieved when the length of PN codes is sufficiently large. However, the upper bounds given by the worst case can also depict the BER performance. The BER performance deteriorates as the number of users increases, but still achieves 10^{-4} under a relatively low symbol SNR, which indicates that the PSS-PPM method can provide communication ability for laser radars.

Fig. 5 shows the root mean square error (RMSE) $\varepsilon_{\hat{D}}$ for distance sensing with respect to the number of users. Due to noise and interference, RMSE only converges in the high-SNR region, which is defined as the asymptotic region. The minimum SNR for convergence and the asymptotic value of RMSE both increase as the number of users increases. However, the performance of MLE is mainly constrained by the sampled resolution, i.e., $\Delta D_s = 7.5$ cm, whose influence can be modeled as a random variable uniformly distributed on $[-\Delta D_s, \Delta D_s]$, so that $\varepsilon_{\hat{D}} \geq \Delta D_s / \sqrt{3} = 4.33$ cm. The quadratic fitting method can break the limitation and its RMSE can approach CRB in the asymptotic region. Moreover, a comparison is made with the single-pulse scheme which is widely adopted by conventional laser radars. The results indicate that the PSS-PPM method provides a sensing ability similar to that of the single-pulse scheme, so it is flexible to not only ISAC scenarios but also conventional sensing tasks.

V. CONCLUSION

The proposed PSS-PPM method provides communication and sensing abilities for the ISAC system. The unipolar

ISAC signal is generated by modulating the position of pulse sequences in each slot so that the IM/DD scheme can be adopted. Correlation-based methods are utilized by both communication and sensing receivers to demodulate information and measure target distance, respectively. Theoretical analysis and simulation results show that communication and sensing can be carried out simultaneously, even in the multi-user scenario. By adjusting the system parameters, the PSS-PPM method can be adopted by different optical ISAC systems, which will play an important role in future optical wireless networks.

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